ParkiStep Twin — Phase 0 Architecture

Forecast FoG onset from clinical context + past-only wearable movement windows

STREAM A | STATIC / HISTORICAL

FoG-STAR clinical_data.csv

Demographics + available clinical assessments

One record per participant; no invented labs/genetics

STREAM B | DYNAMIC / WEARABLE

FoG-STAR sensor_data.csv

4 IMUs • accelerometer + gyroscope • 60 Hz

Replay in timestamp order as simulated live data

Clinical feature pipeline

Validate fields • encode/scale in training folds

Link by subjectID; track missingness

Causal sensor-window pipeline

Validate timestamps / gaps • trailing ~2 s window

Extract features using samples available at time t

FEATURE-LEVEL FUSION

Clinical profile + current sensor features → one prediction row

EARLY-WARNING MODEL

Primary candidate: P(FoG onset in the next 3 seconds)

1 s and 5 s are sensitivity analyses, not promised warning times

DIGITAL TWIN STATE + ALERT MANAGER

Profile • latest window • risk • signal quality • alert history

CLINICIAN DASHBOARD

Patient profile • signal replay • forecast • evidence

Predicted and observed events clearly distinguished

POST-HOC EVALUATION ONLY

Expert FoG annotations compare against alerts

Labels must never be predictor inputs at inference time

Research prototype only • not a diagnostic or treatment system